

MACHINE LEARNING PROJECT

READ ME FILE

Project Overview

This project is all about risk prediction with respect to the loan default considering real-world LendingClub Data.

The two steps taken for the approach are:-

Classification – to predict if a borrower defaults or not.

Regression – to predict how much money could be lost in the event of a default.

These two were integrated in order to give a definition of Expected Financial Loss, which allows bank lenders to know the default's possibility and the impact.

What We Used in our project?

Data: Accepted loan dataset of LendingClub between 2007 and 2018

Libraries: pandas, numpy, scikit-learn, matplotlib, seaborn

Models:

- RandomForestClassifier is used for classification.
- LinearRegression is used to predict loss amount.

The Working of our Project

Here is our code functioning step by step in short:

1. Importing and cleaning the data: No null values and convert text to numeric.
2. New credit score for loan grade-derived features.
3. Data set splitting into training and test data sets.
4. Feature scaling for better model performance.
5. Train-a-classifier for estimating the probability of default.
6. Train a regression model (only on defaulters) to find out how much will be lost.
7. Expected Loss for each borrower.
8. Shows how expected loss vary as per income and loan amount.
9. Performance metrics for both models are printed.

Contribution of our Team members

- Nimmakayala Charitha [SE23UCSE124] - Helped in major parts of the code and also helped in the read me file.
- Nikshipta Reddy [SE23UECM029] - Helped in major part of report and also helped in code.
- Nidhi Reddy [SE23UCSE123] - Helped in major part of read me file and also helped in code.
- Pallavi Katragadda [SE23UCSE123] - Helped in read me file and code.
- Pranishk Reddy [SE23UARI099] - Helped in report and code.